

Computational and Conceptual Questions & Answers

Conceptual Questions

Q: What is a Language Model (LM)?

A: A Language Model is a probabilistic model that learns the likelihood of a sequence of words and predicts the next word in a sentence.

Q: Why is attention important in Transformer models?

A: Attention allows each token to weigh the importance of all other tokens in the input sequence, helping the model understand context and relationships.

Q: What does instruction fine-tuning do in large language models?

A: It refines a pre-trained model using supervised learning with human-annotated inputs and outputs to make the model follow instructions better.

Q: What is the difference between instruction fine-tuning and RLHF?

A: Instruction fine-tuning uses labeled datasets to train, while RLHF adjusts model outputs based on human feedback using reinforcement learning.

Q: How do token embeddings work?

A: Token embeddings convert text tokens into high-dimensional vectors that capture semantic meaning and are refined during model training.

Computational Questions

Q: How is a token embedding computed in a Transformer?

A: It is looked up in an embedding matrix where each token has a corresponding learned vector.

Q: What is the formula for scaled dot-product attention?

A: $\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k})V$, where Q , K , and V are the query, key, and value matrices.

Q: How does the model calculate attention scores?

A: By computing the dot product of the query and key vectors, then scaling and applying a softmax

function.

Q: What are multi-head attention mechanisms?

A: They allow the model to focus on different parts of the input simultaneously by computing attention in multiple subspaces.

Q: How does positional encoding help the Transformer model?

A: It injects information about the position of each token in the sequence since the model architecture is order-invariant.